Plan B: Operating Characteristics of Expectation-Maximization Model Selection for QTL Mapping

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Abstract

Missing data remains a persistent challenge across fields such as genetics, epidemiology, and social sciences. One critical yet under-addressed issue in the treatment of incomplete data is double dipping, the inappropriate reuse of data in both model selection and inference (Jiang, Nguyen, and Rao 2015, 2011). This work investigates how double dipping manifests in the context of missing data imputation and demonstrates that the Expectation-Maximization-Model Selection (E-MS) framework (Jiang, Nguyen, and Rao 2015) effectively addresses aspects of this problem. We compare E-MS to conventional methods across a range of data-generating scenarios under different missing data mechanisms, including missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR). Large-scale simulations are conducted to assess the robustness of E-MS and to evaluate its performance relative to existing approaches. Results demonstrate that E-MS mitigates key biases associated with double dipping, consistently outperforming conventional methods by achieving superior F1 scores through high recall coupled with strong precision.

1 Introduction

In applied statistical analysis, missing data is an unavoidable and often complex obstacle. Whether due to non-response in surveys, sensor failure in experiments, or limitations in sequencing technologies in genomics, analysts must frequently address datasets with incomplete observations. The implications of improper handling of missingness are far-reaching, especially when decisions about model structure or inference procedures are based on incomplete information.

This paper focuses on a particular consequence of missing data analysis: double dipping (Jiang, Nguyen, and Rao 2015, 2011). This occurs when the same data is used in multiple stages of model development without proper adjustment, typically in both model selection and inference. Such practices can inflate Type I error rates and lead to misleading conclusions.

The goal of this work is to understand how imputation methods, particularly the E-MS algorithm, interact with and potentially exacerbate or alleviate double dipping.

Prior work has shown that standard information criteria applied directly to incomplete logistic regression data can systematically favor underspecified models. For example, (Ibrahim, Zhu, and Tang 2008) demonstrate that, under a missing at random mechanism, the intercept-only model may maximize the conditional expected log-likelihood even when it is not the true model. This phenomenon can lead traditional Bayesian Information Criterion (BIC)-based selection procedures, such as the BIC penalty denoted by IZT, to consistently choose an incorrect model. Through simulations across different sample sizes, true model structures, and both ignorable and nonignorable missingness mechanisms, they show that alternative methods like E-MS achieve substantially higher true positive rates and are more robust to the missingness mechanism (Ibrahim, Zhu, and Tang 2008; Rubin 1976).

2 Missing Data

To understand the challenges of missing data, we must first establish notation. Let $Y = (Y_{\mathrm{o}}, Y_{\mathrm{m}})$ denote the complete data matrix, where Y_{o} represents the observed values and Y_{m} the missing values. Define the missingness indicator $R_i \in \{0, 1\}$, where $R_i = 1$ if observation i is observed and $R_i = 0$ if it is missing, with parameter ϕ governing the missingness mechanism and θ governing the data-generating process. The joint distribution then factorizes as:

$$P(Y, R \mid \theta, \phi) = P(Y \mid \theta) \cdot P(R \mid Y, \phi)$$

The three missing data mechanisms, introduced by Rubin (1976), are distinguished by the structure of $P(R \mid Y, \phi)$ and form a strict hierarchy: MCAR $\subset$ MAR $\subset$ MNAR.

Missing at Random (MAR) refers to a situation where the probability of missing data

depends only on the observed data and not on the unobserved (missing) data. Formally:

$$P(R \mid Y, \phi) = P(R \mid Y_{\text{o}}, \phi)$$

In other words, once we account for the observed variables, any remaining missingness is random with respect to the missing values themselves. This allows for unbiased estimation if the missingness is properly accounted for in the analysis. On the other hand, Missing Not at Random (MNAR) occurs when the probability of missing data depends on the unobserved data itself, even after accounting for the observed data:

$$P(R \mid Y, \phi) = P(R \mid Y_{\text{o}}, Y_{\text{m}}, \phi)$$

where the dependence on Y_{m} cannot be reduced to a function of Y_{o} alone. This introduces potential bias because the missingness is directly related to the values of the missing data, making it more challenging to handle in statistical analyses without making strong assumptions about the missing data mechanism. Understanding whether data are MAR or MNAR is crucial for applying appropriate statistical methods and drawing valid conclusions from incomplete datasets.

While Missing Completely at Random (MCAR) would be a special case where missingness is entirely independent of both observed and unobserved data:

$$P(R \mid Y, \phi) = P(R \mid \phi), \quad \text{i.e.,} \quad R \perp\!\!\!\perp Y$$

this assumption is rarely met in real-world data. The three mechanisms are therefore nested, with MCAR being the most restrictive special case and MNAR the most general:

$$\text{MCAR} \implies \text{MAR} \implies \text{MNAR}$$

Incomplete cases, particularly those with only 20–40% of the full information available, often appear in genetic studies such as Quantitative Trait Loci (QTL) mapping, where limitations in data collection or sequencing coverage lead to substantial gaps. While several approaches exist to handle such missing data, ranging from complete-case analysis to multiple imputation, few adequately address the implications of using imputed data during both model selection and final inference.

3 Imputation

Single imputation methods replace each missing value with one deterministic estimate. Although these approaches are easy to implement and computationally efficient, they often fail to capture the uncertainty inherent in missing data. One of the most widely used approaches is mean, median, or mode imputation, in which a missing value is replaced with a measure of central tendency derived from the observed data. While this method preserves sample size and is straightforward to apply, it artificially reduces variability and can bias parameter estimates, especially when the data are not missing completely at random (Little and Rubin 2002).

Another form of single imputation is regression imputation, where missing values are predicted using a regression model built from the observed cases. Although regression-based approaches provide more plausible values than simple mean imputation, they still tend to overstate the precision of the data by treating predicted values as known quantities (Allison 2011).

Multiple imputation, introduced by (Rubin 1987), is widely considered the gold standard for handling missing data in many applied research contexts. Instead of substituting missing values with a single estimate, MI generates multiple versions of the dataset, each containing different plausible imputations drawn from a predictive distribution that reflects both observed data and uncertainty.

After statistical analyses are performed separately on each imputed dataset, the results are combined using Rubin's rules, which account for both within-imputation and between-imputation variability. This leads to unbiased parameter estimates and valid standard errors when data are missing at random (MAR) (Van Buuren 2018). Although MI requires more computation and careful model specification than single imputation, its ability to incorporate uncertainty makes it far more statistically robust.

Model-based imputation methods leverage statistical or machine learning models to estimate missing values in a data-driven manner. These approaches are often more flexible than traditional single or multiple imputation.

A foundational model-based method is the Expectation–Maximization (EM) algorithm, which iteratively estimates missing data by maximizing the likelihood of the observed data under assumed model parameters (Dempster, Laird, and Rubin 1977). EM produces maximum likelihood estimates but does not generate multiple datasets, so it does not account for imputation uncertainty unless paired with MI.

Machine-learning–oriented methods have become increasingly popular. k-Nearest Neighbors (kNN) imputation identifies cases with similar feature profiles and uses their observed values to fill in missing entries. Random forest–based methods, such as missForest, build ensembles of decision trees to predict missing values; they can handle nonlinear relationships and mixed data types effectively (Stekhoven and Bühlmann 2012). More recently, deep learning autoencoder imputation has been used to reconstruct missing values by learning compressed representations of the full dataset, making it particularly effective in high-dimensional or complex data structures.

Model-based imputation techniques offer strong predictive performance and flexibility, especially in modern large-scale datasets. However, they require careful tuning and may not inherently provide valid measures of uncertainty unless extended within a multiple-imputation framework.

4 Double Dipping Problem

In high-dimensional QTL studies and related genetic mapping contexts, a common analytical workflow uses the same dataset both to select models and to estimate their parameters or significance. This practice, often referred to as double dipping or circular analysis, arises when data are used first to generate hypotheses or select features and then reused to estimate or test those features without accounting for the selection process. The statistical danger of this workflow has been well documented in other fields, particularly in neuroimaging, where it was shown that re-analyzing data after selection leads to systematically biased effect sizes and invalid inferential statements (Kriegeskorte et al. (2009)). QTL mapping is a particularly acute instance of this problem. Because the number of candidate loci vastly exceeds the number of observations, researchers must scan across the genome and select regions that exceed some significance threshold — yet most classical statistical inference assumes that the model was specified independently of the data used to estimate it. In QTL mapping, this assumption is routinely violated and the impact is magnified by the scale, sparsity, and missingness that characterize such data.

Markers, covariates, and interaction structures are typically chosen using data-dependent criteria such as likelihood scans, penalized regression, or information criteria, and the same data are then used to estimate effect sizes and uncertainties for the selected loci. Conditioning on the outcome in this way induces selection bias: the chosen loci are those that happened to align with random fluctuations, imputation artifacts, or sampling noise, and treating the selected model as fixed ignores this conditioning. As a result, estimated genetic effects are biased away from zero and uncertainty is understated, even when model selection is performed using principled criteria rather than hypothesis testing (Pasaniuc and Price (2017)).

This issue becomes more severe in the presence of missing data and genotype imputation, which are now unavoidable in modern QTL studies. High-resolution mapping pipelines rely on probabilistic reconstruction of unobserved genotypes from low-coverage sequencing or from related individuals, often in complex multi-parent populations (Huang et al. (2012); Karl

W. Broman et al. (2017)). These imputation models are trained on the observed data and encode assumptions about haplotype structure, recombination, and linkage disequilibrium. When the imputed values are then treated as fixed inputs to downstream QTL scans, the uncertainty and model dependence introduced during imputation are effectively ignored. If marker selection or model fitting is carried out on these imputed datasets, the selection step itself is influenced by the imputation model, which further entangles model choice with the observed data and amplifies double-dipping bias. In this sense, sophisticated imputation does not remove the inferential problem; it relocates it into a more complex, and often hidden, modeling layer.

The practical response to multiple testing in QTL studies is rarely to rely on strict p-value thresholds alone, because the number of candidate loci is enormous and effect sizes are typically small. Instead, researchers often adopt penalized likelihoods, Bayesian model probabilities, or criteria such as BIC to balance goodness-of-fit against model complexity. While these tools are well suited for high-dimensional search, they do not by themselves resolve the inferential consequences of data-dependent model choice. Once a model is selected, standard errors, posterior summaries, or test statistics are usually computed as if the model were fixed in advance, even though it was chosen precisely because it fit the data unusually well. This creates a false sense of certainty about the importance of specific loci, and it is one of the main drivers of the winner's-curse and reproducibility problems in genetic association studies.

A concrete illustration of how information criteria can fail under missing data and data-dependent model selection is given by the following example (Ibrahim, Zhu, and Tang 2008).

Suppose that one attempts to select a logistic model, $\text{logit}(\pi_i) = \mathbf{x}_i^\top \boldsymbol{\beta}$, where $\pi_i = P(Y_i = 1)$, $Y_1, \ldots, Y_n$ being independent binary observations and $\mathbf{x}_i$ is a vector of covariates to be selected. Suppose that $y_1, \ldots, y_5$ are observed and the rest of the y_i's are missing. For simplicity, assume that all the $\mathbf{x}_i$'s are observed and that missingness is MAR (Rubin 1976)

for simplicity.

Let M_0 denote the intercept-only model and suppose that the true model is not M_0. The complete-data log-likelihood under M_0 is $\ell = \sum_{i=1}^{n}\Big\{y_i \log p_0 + (1-y_i)\log(1-p_0)\Big\}$, where $p_0 = e^{\beta_0}/(1+e^{\beta_0})$ and β_0 is the intercept. Under M_0 the conditional expectation given the observed data is $E(\ell \mid y_1, \ldots, y_5, \text{all } \mathbf{x}_i) = \sum_{i=1}^{5}\Big\{y_i \log p_0 + (1-y_i)\log(1-p_0)\Big\} + (n-5)\Big\{p_0 \log p_0 + (1-p_0)\log(1-p_0)\Big\}$. If $y_i = 1$ for $1 \le i \le 5$, then as $p_0 \to 1$ the expression above tends to 0. On the other hand, under any competing model M the corresponding log-likelihood satisfies $\ell = \sum_{i=1}^{n}\{y_i \log p_i + (1-y_i)\log(1-p_i)\} \le 0$, so $E(\ell \mid y_1, \ldots, y_5, \text{all } \mathbf{x}_i) \le 0$ under M.

Therefore the maximized conditional expectation of ℓ under M_0 (which is 0) is greater than or equal to that under any M. Because M_0 also has the smallest dimension, every information criterion of the form considered by (Ibrahim, Zhu, and Tang 2008) will select M_0, the intercept-only model, even though it is known to be incorrect. This shows that principled information criteria can still systematically favor the wrong model whenever missing data create a degenerate likelihood surface for the simplest candidate.

Recent methodological surveys in high-dimensional biomedical inference emphasize that many modern pipelines successfully address feature selection, prediction, and missingness, yet still leave unresolved the deeper statistical issue of how to perform valid inference after selection (Serghiou et al. (2023)). In QTL mapping, where model search, imputation, and inference are tightly interwoven, this gap is particularly consequential. Double dipping is therefore not a technical detail about p-values versus information criteria; it is a structural problem that arises whenever the same data are used to decide what model to believe and to quantify how strongly that model is supported. Without accounting for this dependency, QTL analyses can produce inflated effect estimates, underestimated uncertainty, and spurious discoveries, even when every individual modeling step appears statistically sound.

5 Existing Approaches

5.1 QTL

Quantitative trait locus (QTL) mapping seeks to identify genomic regions that are associated with variation in a quantitative phenotype. In practice, QTL analysis must account for missing or unobserved genotype information between genetic markers, which arises naturally due to limited marker density. The `R/qtl` framework, as developed and documented by (Karl W. Broman et al. 2019) and the `rqtl` manual, provides a comprehensive set of tools for addressing this problem using likelihood- and regression-based approaches.

To identify markers of interest, QTL mapping is commonly performed using three complementary methods: expectation–maximization (EM) algorithms (Dempster, Laird, and Rubin 1977; Lander and Botstein 1989), Haley–Knott (HK) regression (Haley and Knott 1992), and multiple imputation approaches (Sugiyama et al. 2001). All three methods rely on the same underlying genetic model but differ in how they handle uncertainty in the unobserved genotypes at putative QTL positions.

The EM-based approach treats the underlying QTL genotypes as latent variables. At each genomic position, the E-step computes the conditional probabilities of the possible QTL genotypes given the observed marker data and current parameter estimates, while the M-step maximizes the expected complete-data log-likelihood to update model parameters. This method provides statistically efficient inference but can be computationally intensive, particularly for large genomes or complex crosses.

Haley–Knott regression offers a computationally simpler approximation to the EM algorithm. Instead of fully iterating between expectation and maximization, HK regression replaces the unobserved QTL genotypes with their conditional expectations and fits a standard linear regression model. While this approximation leads to substantial gains in speed, it may result in a modest loss of efficiency compared to the full EM approach, especially in regions with sparse marker information.

Multiple imputation methods take a different perspective by repeatedly sampling complete genotype datasets from the conditional distribution of the missing genotypes given the observed markers. For each imputed dataset, a standard QTL scan is performed, and the results are combined across imputations. This strategy more fully propagates uncertainty in the missing genotypes and often yields more robust inference, at the cost of increased computational effort.

In genomic applications, QTL mapping is typically conducted on large-scale datasets involving thousands of genetic markers and, in modern studies, potentially thousands of individuals. In such settings, even small systematic errors in the handling or imputation of missing genotype data can accumulate and substantially bias downstream inference (Fan 2008). Poorly specified imputation models may distort conditional genotype probabilities, inflate or attenuate LOD scores, and lead to spurious QTL detection or missed true signals. Consequently, careful treatment of missing data is critical, as incorrect imputation can greatly skew results despite large sample sizes that might otherwise suggest statistical robustness.

For each mapping method, a genome-wide scan yields a profile of logarithm of the odds (LOD) scores that measure the strength of evidence for the presence of a QTL at each genomic position. Statistical significance is assessed by applying a predefined LOD threshold to these scan results, and all genomic locations where the LOD score exceeds this threshold are identified as candidate QTL. Reporting all detected peaks across the genome for each method enables a systematic comparison of the number, location, and strength of QTL signals, facilitating evaluation of detection consistency and relative mapping performance among the EM, Haley–Knott, and multiple imputation approaches.

5.2 EM - The Expectation-Maximization

The Expectation–Maximization (EM) algorithm, introduced by (Dempster, Laird, and Rubin 1977), provides a likelihood-based framework for parameter estimation in the presence of missing or latent data. Its appeal lies in its conceptual simplicity and broad applicability

across a wide range of statistical models. Let Y_o denote the observed data, Y_m the missing or latent data, and θ the vector of unknown parameters. If the complete data $Y = (Y_o, Y_m)$ were fully observed, inference would proceed by maximizing the complete-data log-likelihood $\ell_c(\theta) = \log p(Y_o, Y_m \mid \theta)$. Because Y_m is unobserved, direct maximization of the observed-data likelihood $\ell(\theta) = \log p(Y_o \mid \theta)$ is often analytically intractable. EM addresses this difficulty by iteratively maximizing a surrogate objective function that is easier to handle.

At iteration t, the E-step constructs the function

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{Y_m \mid Y_o, \theta^{(t)}} \Big[\log p(Y_o, Y_m \mid \theta) \Big],$$

which is the conditional expectation of the complete-data log-likelihood with respect to the distribution of the missing data given the observed data and the current parameter estimates $\theta^{(t)}$. This step does not impute missing values deterministically; instead, it averages over their conditional distribution under the current model, thereby producing an expected log-likelihood that reflects the uncertainty induced by missingness, but only through expectation rather than full probabilistic propagation.

The M-step then updates the parameter estimates by maximizing this expected log-likelihood,

$$\theta^{(t+1)} = \arg\max_{\theta} Q(\theta \mid \theta^{(t)}).$$

This optimization yields new parameter values that increase, or at least do not decrease, the observed-data likelihood. Repeating the E-step and M-step generates a sequence $\theta^{(t)}$ with the property that $\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)})$, and the algorithm is guaranteed to converge to a stationary point of the likelihood under mild regularity conditions.

Despite its strengths, EM has several important limitations. First, it assumes that the model specified for the complete data is correct; any model misspecification directly affects both the conditional distribution used in the E-step and the optimization performed in the M-step. Second, although the E-step computes expectations over the missing data, EM

ultimately treats these expectations as fixed quantities in the M-step, and therefore does not fully propagate uncertainty in the latent variables into parameter uncertainty. Third, the likelihood surface in models with latent structure is often multimodal, and EM may converge to a local rather than global maximum depending on initialization.

6 E-MS

The E–M algorithm is well known for parameter estimation in the presence of missing data. On the other hand, model selection, as another component of model identification, may also be viewed as parameter estimation, with the parameter being the (identification (ID) number of) the model and the parameter space being the (ID numbers of the) model space. Namely, we combine the parameters with the model under which the parameters are defined. So, at the current stage of the iteration, we have the current model, M_c, as well as the current estimates of the parameters, $\hat{\theta}_c$, under M_c. Let $Q(M) = Q(Y, M, \theta_M)$ be a measure of lack-of-fit, where Y represents the complete data, M a candidate model, and θ_M the vector of parameters under M. We take the conditional expectation of $Q(M)$ under M_c, with the parameters under M_c, $\theta_{M_c} \equiv \hat{\theta}_c$, given the observed data y_o, denoted by $\mathbb{E}_c\Big[Q(M) \mid y_o\Big]$. This is the E-step.

In the next step, we carry out model selection using $\mathbb{E}_c\{Q(M) \mid y_o\}$ as the measure of lack-of-fit. To do so, we first find $\hat{Q}_c(M) = \inf_{\theta_M \in \Theta_M} \mathbb{E}_c\{Q(M) \mid y_o\}$, where Θ_M is the parameter space under M. We can use $\hat{Q}_c(M)$ in a GIC setting, in which the optimal model, $\hat{M}_{\text{opt}}$, is found by minimizing $\hat{Q}_c(M) + \lambda_n|M|$ over $M \in \mathcal{M}$, the class of candidate models, where λ_n is a penalty that depends on the sample size n, and $|M|$ is the dimension of M. Alternatively, we may use the fence method (Jiang et al. 2008) based on $\hat{Q}_c(M)$. This is the MS-step, where MS stands for "model selection." We then replace M_c by $\hat{M}_{\text{opt}}$, found in the MS-step, and $\hat{\theta}_c$ by $\hat{\theta}_{\text{opt}}$, where $\hat{\theta}_{\text{opt}}$ is the parameter vector under $\hat{M}_{\text{opt}}$ corresponding to the minimizer of $\mathbb{E}_c\{Q(\hat{M}_{\text{opt}}) \mid y_o\}$ over $\theta_{M_{\text{opt}}} \in \Theta_{\hat{M}_{\text{opt}}}$, and return to the E-step. We illustrate

the EM–MS procedure with some examples.

Following (Karl W. Broman and Speed 2002), we have a conditional linear regression model for the phenotype variable Y_i such that, given the marker indicators x_i, we have $Y_i = \sum_{k=1}^{r} \sum_{j \in M_k} \beta_{kj} X_{ijk} + \varepsilon_i$, where r is the number of chromosomes, M_k is a subset of $\{1, \ldots, q\}$, and q is the number of markers on each chromosome, and ε_i is a normal error with mean zero and unknown variance σ^2. The ε_i's are uncorrelated and also independent with the X_{ijk}'s. Furthermore, the marker indicators X_{ijk} are assumed to be a Markov chain within each chromosome, with $P(X_{i1k} = 0) = P(X_{i1k} = 1) = \frac{1}{2}$ (Mendel's rule), and $P(X_{i,j+1,k} = 1 \mid X_{ijk} = 0) = P(X_{i,j+1,k} = 0 \mid X_{ijk} = 1) = \theta$, where θ is the recombination fraction. The problem of interest is to identify the subset $M = (M_1, \ldots, M_r)$, which is viewed as a model selection problem as in (Karl W. Broman and Speed 2002).

We consider the E–MS in conjunction with the BIC procedure. Due to the high dimensionality, we consider the forward/backward (FW/BW) BIC procedure of (Karl W. Broman and Speed 2002). A detailed description of the latter is given in the online supplemental material (Section A.3). The log-likelihood, under a given model M, can be expressed as $\ell_M = \ell_{M,y|x} + \ell_x$, where ℓ_x does not depend on the model. Here, $\ell_{M,y|x} = -0.5(n \log \sigma^2 + \sigma^{-2} \sum_{i=1}^{n} (y_i - x_i' \beta_M)^2)$, where c is a constant. Thus, we have $\mathrm{BIC}(M) = -2\ell_M + |M| \log(n)$, where $\hat{\ell}_M$ is the maximized ℓ_M with respect to M and the parameters. It is easy to show that the MLE of θ, $\hat{\theta}$, is the same as the maximizer of ℓ_x, which does not depend on M. Thus, we have ...

$$\mathrm{BIC}(M) = -2\hat{\ell}_{M,\,y|x} + |M| \log(n) - 2\hat{\ell}_x \propto -2\hat{\ell}_{M,\,y|x} + |M| \log(n). \tag{1}$$

In addition, the FW/BW requires evaluation of $\min_\beta \mathrm{RSS}(y, X, \beta)$, where

$$\mathrm{RSS}(y, X, \beta) = \sum_{i=1}^{n} (y_i - x_i' \beta)^2. \tag{2}$$

with $X = (x_i')_{1 \le i \le n}$. Because both (1) and (2) involve missing data, we replace them by their conditional expectations under the current model M_c and the current parameter

estimates under M_c, before the minimization/maximization. This leads to

$$\mathrm{RSS}_c(M \mid y_o, x_o) = \min_{\beta} \mathbb{E}_c[\mathrm{RSS}(Y, X, \beta) \mid y_o, x_o]\,, \tag{3}$$

$$\mathrm{BIC}_c(M \mid y_o, x_o) = -2 \max_{\beta,\sigma^2} \mathbb{E}_c[\ell(M, Y \mid X) \mid y_o, x_o] + |M| \log(n). \tag{4}$$

Both have closed-form expressions, where x_o and y_o denote the observed x's and y's, respectively.

In summary, given M_c and the current parameter estimates, the FW/BW, based on (3), is used to generate a sequence of models; the BIC, based on (4), is then applied to the sequence generated by the FW/BW to update the model as well as parameter estimates.

A reasonable initial model is the full model, M_f. A reasonable initial estimator for θ is $\hat{\theta}_0$, the proportion of observed cases in which $x_{i,j+1,k}$ and x_{ijk} are different. As for the initial estimator of $\hat{\beta}_f$, the vector of regression coefficients under M_f, note that the idea of least squares (LS) fit in regression is to find the parameter estimates that minimize $\sum_{i=1}^n (y_i - \mathbb{E}_f(Y_i \mid x_o))^2$, where $\mathbb{E}_f$ denotes expectation under M_f. Due to the missing data, it is natural to replace this: $\sum_{i=1}^n (y_i - \mathbb{E}_f(Y_i \mid x_i))^2$ by $\sum_{i \in I_o} (y_i - \mathbb{E}_f(Y_i \mid x_{i,o}))^2$, where I_o denotes the subset of indices i such that y_i is observed. Furthermore, we have $\mathbb{E}_f(Y_i \mid x_o) = \sum_{k=1}^r \sum_{j=1}^q \beta_{jk}\, \mathbb{E}_f(X_{ijk} \mid x_{o,i})$, where $x_{i,o}$ denotes the observed x's for the i-th subject, and an expression of the conditional expectation can be easily obtained, with $\hat{\theta}_0$ replacing θ, if x_{ijk} is missing. We then run the LS with y_i, $i \in I_o$, as the responses and $\mathbb{E}_f(X_{ijk} \mid x_{o,i})$, $j = 1, \ldots, q$, as the predictors, to obtain the initial estimator $\hat{\beta}_{f,0}$ for β_f. The initial estimator for $\sigma^2, \hat{\sigma_0^2}$, is the RSS of this LS fit divided by $|I_o| - qr$.

Let y, x, x_c, x_d denote the data for $y_i, x_i, x_{i,c}, x_{i,d}$, respectively, across $1 \le i \le n$. Then, it can be shown that the complete-data log-likelihood has the expression

$$\ell = c - \frac{n}{2}(\log\sigma^2 + \log|\Omega|) + \sum_{r=1}^{s} n_r \log\pi_r$$
$$-\frac{1}{2}\sum_{r=1}^{s}\sum_{i=1}^{n} \mathbf{1}_{(x_{i,d}=v_r)}(x_{i,c}-\mu_r)'\Omega^{-1}(x_{i,c}-\mu_r) \tag{5}$$
$$-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - x'_{i,M}\beta_M)^2,$$

where c is a constant. Note that the maximum likelihood is a constrained maximization problem, namely, maximize ℓ subject to $\sum_{r=1}^{s}\pi_r = 1$. Define $\mathcal{L} = \ell + \lambda\left(\sum_{r=1}^{s}\pi_r - 1\right)$. Then, the MLE of the parameters, plus the Lagrange multiplier λ, is a stationary point of $\mathcal{L}$.

To see an alternative, we now consider the E–MS in conjunction with the adaptive fence (Jiang et al. 2008). See (Jiang 2014) for a recent review of the fence methods. Take the initial model, M_0, as the full model, M_f, and let β_f be β under M_f. Let $\mathbb{E}_f$ denote the conditional expectation under M_f and the current estimates of parameters under M_f, including β_f, σ^2, μ_r, π_r, $r = 1 \le r \le s$, and Ω. Let y_o, x_o denote the observed y, x, respectively. By (5), with $M = M_f$, we have

$$\bar{\ell}_f = \mathbb{E}_f(\ell \mid y_o, x_o) = c - \frac{n}{2}\log\sigma^2$$
$$-\frac{1}{2\sigma^2}\sum_{i=1}^{n}\mathbb{E}_f\left[(y_i - X_i'\beta_f)^2 \mid y_o, x_o\right], \tag{6}$$

where c does not depend on β_f and σ^2. From (6), we obtain the updates for β_f and σ^2,

$$\hat{\beta}_f = S_2^{-1}S_1, \qquad \hat{\sigma}^2 = n^{-1}\Big(S_0 - S_1'S_2^{-1}S_1\Big). \tag{7}$$

The quantities in (7) are

$S_0 = \sum_{i=1}^{n}\mathbb{E}_f(Y_i^2 \mid y_o, x_o)$, $S_1 = \sum_{i=1}^{n}\mathbb{E}_f(X_iY_i \mid y_o, x_o)$, $S_2 = \sum_{i=1}^{n}\mathbb{E}_f(X_iX_i' \mid y_o, x_o)$. Furthermore, we have (see Section A.4.2 of the online supplementary material from (Jiang, Nguyen, and Rao 2015))

$$\begin{aligned}\hat{\mu}_r &= \frac{\sum_{i=1}^n \mathbb{E}_f\left[\mathbf{1}_{(X_{i,d}=v_r)}\, X_{i,c} \mid y_o, x_o\right]}{\sum_{i=1}^n P_f(X_{i,d}=v_r \mid y_o, x_o)},\\ \hat{\pi}_r &= \frac{\mathbb{E}_f(N_r \mid y_o, x_o)}{\sum_{r=1}^s \mathbb{E}_f(N_r \mid y_o, x_o)}, \qquad 1 \le r \le s.\end{aligned} \tag{8}$$

$$\hat{\Omega} = \frac{1}{n}\sum_{r=1}^{s}\sum_{i=1}^{n} \mathbb{E}_f\left[\mathbf{1}_{(X_{i,d}=v_r)}\,(X_{i,c}-\hat{\mu}_r)\,(X_{i,c}-\hat{\mu}_r)^\top \;\middle|\; y_o, x_o\right]. \tag{9}$$

It remains to evaluate the conditional expectations involved in (7)–(9). Let y_m, x_m, x_c, x_d denote the missing parts of y, x, x_c, and x_d, respectively. Although it is possible to obtain a conditional density $f(y_m, x_m \mid y_o, x_o)$, the result is not a common distribution (e.g., normal), under which the conditional expectations can be easily obtained analytically. Alternatively, one may consider sampling from the conditional distribution and use the Monte Carlo method to compute the conditional expectations. To do so, first note that it is easy to show that one can sample from the joint conditional distribution by sampling independently from the conditional distribution for each subject. To sample from the subject conditional distribution, note that $f_{M,i}(y_{i,m}, x_{i,m} \mid y_{i,o}, x_{i,o}) \propto f_{M,i}(y_i, x_i), \propto$

$$\exp\left[\sum_{r=1}^{s}\mathbf{1}_{(x_{i,d}=v_r)}\Big\{\log\pi_r - \frac{1}{2}(x_{i,c}-\mu_r)'\Omega^{-1}(x_{i,c}-\mu_r)\Big\} \qquad - \frac{1}{2\sigma^2}\Big(y_i - x'_{i,M}\beta_M\Big)^2\right].$$

where $\propto$ means that the expression is up to a function of $y_{i,o}, x_{i,o}$, which is considered constant during the sampling of $y_{i,m}, x_{i,m}$. Next, we employ the Metropolized independence sampler (MIS; e.g., (Liu 2004, 115)), which is a special case of the Metropolis–Hastings algorithm. We refer readers to (Jiang, Nguyen, and Rao 2015) Section A.4.2 of the online supplementary material.

The initial estimates of μ_r, $1 \le r \le s$, Ω, π_r, $1 \le r \le s$, are $\mu_r^{(0)} = n_r^{-1}\sum_{i\in I_{r,0}} x_{ir}$, $1 \le r \le s$, where $I_{r,0} = \{i \le n : x_{ir}$ is observed and $x_{i,d} = r\}$, and $n_r = |I_{r,0}|$. Let $\hat{\Omega}^{(0)} = n_0^{-1}\sum_{r=1}^s\sum_{i\in I_{r,0}}(x_{i,c}-\mu_r^{(0)})(x_{i,c}-\mu_r^{(0)})^\top$, where $I_0 = \bigcup_{r=1}^s I_{r,0}$ and $n_0 = |I_0|$, and

$\pi_r^{(0)} = \#\{i \le n : x_{i,d} = r\}/n$, $1 \le r \le s$. Furthermore, the initial estimate of $\boldsymbol{\beta}$ is the least-squares estimate based on the half-observed data, that is, $\hat{\boldsymbol{\beta}}^{(0)} = (X_{ao}^\top X_{ao})^{-1} X_{ao}^\top y_{ao}$ (assuming, without loss of generality, that $X_{ao}^\top X_{ao}$ is nonsingular), where $X_{ao} = (x_i^\top)_{i \in I_{ao}}$, $I_{ao} = \{1 \le i \le n : x_i, y_i \text{ observed}\}$, and $y_{ao} = (y_i)_{i \in I_{ao}}$. The initial estimate of σ^2 is $(\hat{\sigma}^2)^{(0)} = \|y_{ao} - X_{ao}\hat{\boldsymbol{\beta}}^{(0)}\|^2/(|I_{ao}| - p - q)$.

For any candidate model M, let $Q(M) = S_0 - S_1^\top S_2^{-1} S_1$, where S_0 is the same as that below (7), and S_{ij}, $j = 1, 2$, are the same as those below (7) with x_{if} replaced by $x_{i,M}$. Note that the conditional expectation E_c will be done by the conditional sampling method mentioned above, with $M = M_0$. Run the AF procedure with $Q(M)$ being the measure of lack-of-fit. Denote the model selected by AF by $\hat{M}$. Let $\hat{\boldsymbol{\beta}} = S_2^{-1} S_1$, where S_{ij}, $j = 1, 2$, are given below (7) with x_{if} replaced by $x_{i,\hat{M}}$. Next, let $\hat{\sigma}^2$ be given by (7), where S_{ij}, $j = 0, 1, 2$, are given below (7) with x_{if} replaced by $x_{i,\hat{M}}$. Also, let $\hat{\rho}_{f,r}$, $1 \le r \le s$, and $\hat{\pi}_r$, $1 \le f \le s$, be given by (8) and (9) (note that these depend only on $M_0 = M_f$, but not on M).

Replace M_0 by $\hat{M}$, and the initial estimates by $\hat{\boldsymbol{\beta}}$, $\hat{\sigma}^2$, $\hat{\mu}_f$, $1 \le f \le s$, $\hat{\pi}_r$, $1 \le r \le s$, and repeat the process. Note that, after this iteration, the E–step is replaced by $\hat{E}_c$, evaluated by the conditional sampling method with $M = \hat{M}$. Keep updating the model and parameters iteratively until convergence (see below).

The E-MS algorithm with generalized information criteria converges globally under certain regularity conditions proposed by (Jiang, Nguyen, and Rao 2015), meaning the iterative procedure will reach a stable solution from any starting point. Furthermore, as sample size increases, the algorithm consistently selects the optimal true model of minimum dimension (Jiang, Nguyen, and Rao 2015), provided that underfitting models have sufficiently poor fit and overfitting models are adequately penalized.

6.1 Merits, Limitations, and Applicability of E-MS

The E-MS algorithm, introduced by (Jiang, Nguyen, and Rao 2015), is an extension of the Expectation-Maximization (E-M) algorithm aimed at addressing model selection in

the presence of incomplete data. Traditional model selection techniques such as AIC or BIC require the complete-data likelihood, which is not directly accessible when data are missing. The E-MS algorithm integrates model selection into the iterative estimation process by performing two key steps at each iteration. First, in the E-step, it computes the expected log-likelihood given the current parameter estimates and the observed data, just as in the classical E-M approach. Then, instead of simply maximizing this expectation, it enters an MS-step, where it evaluates a set of candidate models using model selection criteria (such as AIC or BIC) applied to the expected log-likelihood. The model with the most favorable score is selected, and parameter estimation continues under that model. This iterative process continues until convergence, producing both parameter estimates and a selected model.

Despite its usefulness, the E-MS algorithm comes with certain limitations. One concern is its computational intensity, as multiple candidate models must be evaluated at each iteration, which can be particularly burdensome in large model spaces. Additionally, like the E-M algorithm, it is sensitive to the choice of initial values and may converge to local optima rather than the global solution. Furthermore, because the selected model may change during the iterations, it introduces complexity in assessing the stability and uncertainty of the final chosen model.

The E-MS algorithm offers several benefits in settings with missing data. It enables principled model selection without requiring prior imputation or complete-case analysis, making it particularly useful in practical applications where data are often incomplete. By embedding model selection directly within the estimation process, it allows for a more integrated and potentially more accurate inference. Moreover, the algorithm is flexible and can be adapted to various modeling contexts, such as regression, variable selection, and latent variable models.

6.2 Purpose of the Work

Although the E-MS algorithm provides a theoretically principled framework for simultaneous model selection and parameter estimation under missing data, its practicality must be validated to recommended it as a viable alternative to conventional methods. Empirical evaluation is therefore essential to determine whether the E-MS algorithm performs reliably as an analytical strategy for QTL mapping in the presence of missing genetic marker data. By directly embedding model selection within the estimation procedure, the proposed approach aims to improve both the accuracy and interpretability of QTL detection without relying on pre-processing steps that may distort the underlying data structure.

To assess whether this approach delivers on its theoretical promise, we conduct a systematic simulation study across a range of sample sizes, noise configurations, and missing data rates, benchmarking E-MS performance against four competing methods. The goal is to determine under which practical conditions E-MS offers a meaningful advantage and where its limitations may warrant caution in applied genomic research.

7 Simulations

We employed a Monte Carlo simulation study to evaluate the performance of the proposed method under varying data conditions. The simulation framework was implemented in R version 4.2.3 and designed with a factorial experimental design to systematically assess method robustness across multiple scenarios. This approach allowed us to examine how the method performs across realistic variations in sample size, measurement error, and data completeness.

The simulation study utilized a full factorial design with three factors to comprehensively evaluate method performance. Sample size was examined at two levels (n = 100 and n = 50) to assess performance in moderate and small sample contexts typical of genetic marker studies. Three distinct noise configurations were implemented to represent different data

quality scenarios: uniform high noise ($\sigma = 1.0$ for all six markers), uniform low noise ($\sigma = 0.1$ for all six markers), and heterogeneous noise ($\sigma = [1.0, 0.1, 0.5, 0.4, 0.3, 0.9]$), representing realistic scenarios where measurement error varies across markers. Missing data rates were evaluated at three levels (5% of markers at random, 1% of markers at random, and naive 10% of rows or individuals removed) to examine method sensitivity to incomplete data, a common challenge in genetic studies. This factorial structure yielded 18 unique experimental conditions ($2 \times 3 \times 3$).

Data generation was performed using the custom function, which simulated genetic marker data across 5 chromosomes with 6 markers per chromosome under the specified noise and missing data conditions. For each of the 18 experimental conditions, we generated 50 independent replicate datasets of size n $\times$ 30 to ensure adequate statistical power for performance assessment, resulting in 900 total simulated datasets. Each replicate was generated using a unique random seed (ranging from 1 to 299) to ensure reproducibility while allowing for replacement of failed simulations due to non-convergence or numerical instability.

To manage the computational demands of this large-scale simulation study, we implemented an adaptive checkpoint-restart system that automatically detected previously completed replicates and resumed from the last successful iteration, preventing redundant computation and allowing the study to be conducted across multiple computing sessions. All simulation errors were logged with timestamps to facilitate debugging and ensure transparency in the data generation process. Simulations that failed to converge within reasonable computational bounds were excluded, and alternative random seeds were used until the target of 50 successful replicates per condition was achieved, with a maximum of 299 attempts permitted per condition to ensure methodological rigor while maintaining computational feasibility.

8 Synthesizing Results

We evaluated five methods: E-MS, BIC, EM, Haley-Knott regression (HK), and imputation-based QTL mapping (IMP) across a combination of sample sizes ($n \in \{50, 100\}$), missingness levels ($\delta \in \{0.01, 0.05, 0.10\}$), and noise structures (strong uniform, weak uniform, random). Performance was assessed using three standard evaluation metrics: precision (the proportion of predicted positives that were true positives), recall (the proportion of actual positives correctly identified by the model), and F1 score (the harmonic mean of precision and recall, providing a single balanced measure of model performance). Full numerical results are reported in Table 1 and 2 and visualized in Figure 1.

Across all simulation conditions at $n = 100$, E-MS consistently demonstrated superior performance relative to all other approaches (Table 1, Figure 2). Key take-home messages are:

- E-MS achieved the best balance of precision and recall, with F1 scores ranging from 0.437 to 0.933, near-perfect recall (0.287–1.0), and high precision (0.835–0.880).
- BIC showed moderate but variable performance, with F1 scores of 0.422–0.950. While BIC achieved high recall (0.640–1.0), its precision was lower (0.320–0.912), reflecting elevated false positive rates relative to E-MS.
- QTL-based methods (EM, HK, IMP) were overly conservative, maintaining perfect precision (1.0) but very low recall (0.287–0.367), resulting in F1 scores of only 0.425–0.531. These methods missed the majority of true QTL signals.

Reducing the sample size from $n = 100$ to $n = 50$ degraded performance across all methods, but to markedly different degrees (Table 2):

- E-MS was the most resilient, with average F1 declining from 0.900 to 0.761 (15.4% reduction).
- BIC was more sensitive to sample size, dropping from an average F1 of 0.807 to 0.657 (18.6% reduction).
- QTL methods suffered the most severe degradation, with average F1 falling from 0.496

to 0.356 (28.2% reduction), highlighting that purely model-selection-based approaches are most vulnerable to limited sample availability.

- E-MS maintained stable performance across missingness levels. At $n = 100$, F1 scores ranged from 0.900–0.923 under low missingness ($\delta = 0.01$), declining modestly to 0.801–0.950 under high missingness ($\delta = 0.10$).
- BIC exhibited greater variability, with F1 scores ranging from 0.422–0.950 at $n = 100$ depending on missingness and noise structure, compared to E-MS's more stable 0.437–0.933 range.
- E-MS was substantially more robust to false positives induced by missingness. Increasing missingness from $\delta = 0.01$ to $\delta = 0.10$ increased average false positives by 31% for BIC but only 18% for E-MS across all noise conditions and sample sizes.
- Under strong uniform noise, all methods performed relatively well at $n = 100$: E-MS achieved F1 scores of 0.921–0.933 and BIC reached 0.785–0.950.
- Under weak uniform noise, E-MS maintained robust performance (F1: 0.437–0.902), while BIC declined to 0.422–0.801 and QTL methods dropped to 0.425–0.474. This represents a 114% performance gap between noise conditions for BIC, compared to only 51% for E-MS, underscoring E-MS's superior ability to detect subtle signals.
- Random noise produced intermediate results, with E-MS F1 scores of 0.489–0.930 at $n = 100$, BIC achieving 0.641–0.883, and QTL methods reaching 0.489–0.526.
- At $n = 50$, noise effects were amplified. Under weak uniform noise with $\delta = 0.05$, E-MS declined from 0.902 to 0.653 F1 (27.6%), while BIC dropped from 0.422 to 0.331 (21.6%, but from a lower baseline). Weak noise combined with small sample size represents the greatest challenge across all methods.

The results collectively establish E-MS as the most robust method across the conditions examined. Its iterative mechanism for jointly updating the selected model and parameter estimates under incomplete data provides consistent advantages in precision-recall balance, resilience to sample size reduction, stability under increasing missingness, and sensitivity to

weak signals. Figure 1 provides a consolidated comparison of all methods across all conditions.

9 Discussion

This work set out to examine the statistical consequences of double dipping in the presence of missing data, and to evaluate whether the E-MS framework of Jiang, Nguyen, and Rao (2015) provides a principled remedy. We began by motivating the problem through the logistic model example of Ibrahim, Zhu, and Tang (2008), which showed that even well-regarded information criteria can systematically select an incorrect model when missing data create a degenerate likelihood surface. This failure is not an artifact of a poorly chosen criterion; it is a structural consequence of reusing the same data for both selection and estimation without accounting for the conditioning that selection induces.

The E-MS algorithm addresses this by embedding model selection directly within the iterative estimation process. Rather than treating the selected model as fixed and then estimating parameters, E-MS alternates between an E-step, which computes the conditional expectation of the complete-data log-likelihood given the observed data and current parameter estimates, and an MS-step, which evaluates candidate models using that conditional expectation as the measure of fit. We reviewed how this procedure applies naturally to the QTL mapping setting of Broman and Speed (2002), where genotype data are partially missing and the model space is defined by subsets of chromosomal markers. Under both BIC-based and adaptive fence-based selection, the E-MS procedure iteratively refines the selected model and parameter estimates until convergence, yielding a joint solution that is consistent with the observed data without the selection bias introduced by double dipping.

A key theoretical result is that the E-MS algorithm converges globally under mild regularity conditions and, as sample size grows, consistently recovers the true model of minimum dimension. This consistency guarantee is meaningful in the missing data context precisely because it holds without requiring complete-case analysis or prior imputation, both

of which can introduce their own inferential distortions. The benefits of E-MS — integrated model selection, flexibility across MCAR, MAR, and MNAR mechanisms, and applicability to regression, variable selection, and latent variable models — are therefore most evident in settings where missingness is substantial and the model space is non-trivial.

This work demonstrates that the range of problems that arise from double dipping in missing data analysis is substantial and multifaceted, ranging from modest sample sizes to large-scale genomic studies. Although the E-MS framework effectively mitigates double dipping bias across diverse missing data mechanisms, its practical application reveals important scaling challenges. Scaling the MS-step in high-dimensional settings requires adopting more robust model selectors beyond standard BIC, as the computational burden and model space complexity grow prohibitively large. To address these limitations and accommodate high-dimensional data, we propose two complementary strategies: (1) multithreading approaches that parallelize model evaluation across candidate models and (2) chunking strategies that decompose the high-dimensional problem into computationally tractable subproblems. These modifications allow practitioners to apply the principles of E-MS to realistic genomic and epidemiological datasets where both missingness and dimensionality present substantial inferential challenges. Future work should focus on developing adaptive threshold criteria for model selection in high dimensions and on implementing efficient computational infrastructure that can scale E-MS to modern biobank-scale datasets.

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Table 1: Results for BIC, EMS, and three QTL mapping approaches (EM, Haley-Knott, and multiple imputation) under different missingness levels (0.01, 0.05, and 0.10) and noise structures (uniform strong, uniform weak, and random) with sample size n=100. Performance is evaluated using true positives, false positives, false negatives, precision, recall, and F1 score. Overall, EMS consistently achieves the highest precision and F1 scores with low false negatives across most scenarios, while BIC shows moderate performance with some false positives. The QTL-based methods generally produce fewer false positives but lower recall, resulting in comparatively lower F1 scores, particularly under weaker or random noise conditions. These results indicate that EMS provides the most balanced and robust performance for the simulated settings.

Method	Missingness	Noise	True Positives	False Positives	False Negatives	Precision	Recall	F1 Score
BIC	0.05	Uniform Strong	6.00	3.70	0.00	0.664	1.000	0.785
EMS	0.05	Uniform Strong	6.00	0.98	0.00	0.877	1.000	0.930
QTL EM	0.05	Uniform Strong	2.20	0.00	3.80	1.000	0.367	0.530
QTL HK	0.05	Uniform Strong	2.18	0.00	3.82	1.000	0.363	0.527
QTL IMP	0.05	Uniform Strong	2.02	0.00	3.98	1.000	0.337	0.496
BIC	0.05	Uniform Weak	3.84	8.60	2.16	0.320	0.640	0.422
EMS	0.05	Uniform Weak	5.68	0.98	0.32	0.870	0.947	0.902
QTL EM	0.05	Uniform Weak	1.78	0.00	4.22	1.000	0.297	0.450
QTL HK	0.05	Uniform Weak	1.76	0.00	4.24	1.000	0.293	0.446
QTL IMP	0.05	Uniform Weak	1.72	0.00	4.28	1.000	0.287	0.437
BIC	0.05	Random	5.30	5.70	0.70	0.520	0.883	0.641
EMS	0.05	Random	5.96	0.92	0.04	0.883	0.993	0.930
QTL EM	0.05	Random	2.16	0.00	3.84	1.000	0.360	0.525
QTL HK	0.05	Random	2.08	0.00	3.92	1.000	0.347	0.510
QTL IMP	0.05	Random	1.98	0.00	4.02	1.000	0.330	0.489
BIC	0.01	Uniform Strong	6.00	1.12	0.00	0.869	1.000	0.923
EMS	0.01	Uniform Strong	6.00	0.92	0.00	0.879	1.000	0.933
QTL EM	0.01	Uniform Strong	2.20	0.00	3.80	1.000	0.367	0.530
QTL HK	0.01	Uniform Strong	2.16	0.00	3.84	1.000	0.360	0.523
QTL IMP	0.01	Uniform Strong	2.10	0.00	3.90	1.000	0.350	0.511
BIC	0.01	Uniform Weak	4.66	2.08	1.34	0.720	0.777	0.739
EMS	0.01	Uniform Weak	5.62	0.90	0.38	0.872	0.937	0.900
QTL EM	0.01	Uniform Weak	1.90	0.00	4.10	1.000	0.317	0.474
QTL HK	0.01	Uniform Weak	1.88	0.00	4.12	1.000	0.313	0.470
QTL IMP	0.01	Uniform Weak	1.84	0.00	4.16	1.000	0.307	0.463
BIC	0.01	Random	5.46	1.04	0.54	0.867	0.910	0.879
EMS	0.01	Random	5.88	0.90	0.12	0.880	0.980	0.923
QTL EM	0.01	Random	2.16	0.00	3.84	1.000	0.360	0.526
QTL HK	0.01	Random	2.10	0.00	3.90	1.000	0.350	0.515
QTL IMP	0.01	Random	2.04	0.00	3.96	1.000	0.340	0.502
BIC	0.10	Uniform Strong	6.00	0.70	0.00	0.912	1.000	0.950
EMS	0.10	Uniform Strong	6.00	1.16	0.00	0.864	1.000	0.921
QTL EM	0.10	Uniform Strong	2.20	0.00	3.80	1.000	0.367	0.531
QTL HK	0.10	Uniform Strong	2.16	0.00	3.84	1.000	0.360	0.525
QTL IMP	0.10	Uniform Strong	2.06	0.00	3.94	1.000	0.343	0.504
BIC	0.10	Uniform Weak	4.90	1.40	1.10	0.803	0.817	0.801
EMS	0.10	Uniform Weak	5.52	1.36	0.48	0.835	0.920	0.867
QTL EM	0.10	Uniform Weak	1.72	0.00	4.28	1.000	0.287	0.438
QTL HK	0.10	Uniform Weak	1.72	0.00	4.28	1.000	0.287	0.438
QTL IMP	0.10	Uniform Weak	1.66	0.00	4.34	1.000	0.277	0.425
BIC	0.10	Random	5.40	0.88	0.60	0.879	0.900	0.883
EMS	0.10	Random	5.80	1.26	0.20	0.851	0.967	0.898
QTL EM	0.10	Random	2.10	0.00	3.90	1.000	0.350	0.513
QTL HK	0.10	Random	2.06	0.00	3.94	1.000	0.343	0.505
QTL IMP	0.10	Random	2.00	0.00	4.00	1.000	0.333	0.493

Table 2: Results for BIC, EMS, and three QTL mapping approaches (EM, Haley-Knott, and multiple imputation) under different missingness levels (0.01, 0.05, and 0.10) and noise structures (uniform strong, uniform weak, and random) with sample size n=50. Performance metrics include true positives, false positives, false negatives, precision, recall, and F1 score. Compared with the larger sample size setting, overall performance decreases, particularly under uniform weak and random noise. EMS generally maintains higher recall and competitive F1 scores across most scenarios, while BIC often identifies more true positives but at the cost of substantially higher false positives, leading to lower precision. The QTL-based methods tend to produce few or no false positives but exhibit lower recall, which reduces their overall F1 scores. These results highlight the impact of smaller sample sizes on variable selection accuracy.

Method	Missingness	Noise	True Positives	False Positives	False Negatives	Precision	Recall	F1 Score
BIC	0.05	Uniform Strong	5.90	4.58	0.10	0.611	0.983	0.737
EMS	0.05	Uniform Strong	6.00	2.52	0.00	0.742	1.000	0.841
QTL EM	0.05	Uniform Strong	1.40	0.00	4.60	0.960	0.233	0.368
QTL HK	0.05	Uniform Strong	1.36	0.00	4.64	0.960	0.227	0.360
QTL IMP	0.05	Uniform Strong	1.32	0.00	4.68	0.960	0.220	0.351
BIC	0.05	Uniform Weak	2.62	7.54	3.38	0.295	0.437	0.331
EMS	0.05	Uniform Weak	4.36	3.08	1.64	0.618	0.727	0.653
QTL EM	0.05	Uniform Weak	1.18	0.00	4.82	0.900	0.197	0.317
QTL HK	0.05	Uniform Weak	1.16	0.00	4.84	0.900	0.193	0.313
QTL IMP	0.05	Uniform Weak	1.08	0.00	4.92	0.840	0.180	0.291
BIC	0.05	Random	4.86	6.00	1.14	0.500	0.810	0.599
EMS	0.05	Random	5.58	2.56	0.42	0.724	0.930	0.802
QTL EM	0.05	Random	1.42	0.00	4.58	0.980	0.237	0.374
QTL HK	0.05	Random	1.40	0.00	4.60	0.980	0.233	0.370
QTL IMP	0.05	Random	1.38	0.00	4.62	0.960	0.230	0.364
BIC	0.01	Uniform Strong	6.00	2.44	0.00	0.749	1.000	0.846
EMS	0.01	Uniform Strong	6.00	2.36	0.00	0.754	1.000	0.850
QTL EM	0.01	Uniform Strong	1.48	0.00	4.52	0.980	0.247	0.387
QTL HK	0.01	Uniform Strong	1.50	0.00	4.50	0.980	0.250	0.390
QTL IMP	0.01	Uniform Strong	1.42	0.00	4.58	0.980	0.237	0.374
BIC	0.01	Uniform Weak	3.56	3.84	2.44	0.529	0.593	0.540
EMS	0.01	Uniform Weak	4.70	2.68	1.30	0.687	0.783	0.718
QTL EM	0.01	Uniform Weak	1.10	0.00	4.90	0.880	0.183	0.299
QTL HK	0.01	Uniform Weak	1.00	0.00	5.00	0.840	0.167	0.274
QTL IMP	0.01	Uniform Weak	0.98	0.00	5.02	0.840	0.163	0.270
BIC	0.01	Random	5.30	3.02	0.70	0.692	0.883	0.759
EMS	0.01	Random	5.68	2.46	0.32	0.740	0.947	0.818
QTL EM	0.01	Random	1.44	0.00	4.56	0.960	0.240	0.377
QTL HK	0.01	Random	1.40	0.00	4.60	0.960	0.233	0.369
QTL IMP	0.01	Random	1.36	0.00	4.64	0.980	0.227	0.361
BIC	0.10	Uniform Strong	5.96	2.26	0.04	0.765	0.993	0.853
EMS	0.10	Uniform Strong	6.00	2.82	0.00	0.720	1.000	0.826
QTL EM	0.10	Uniform Strong	1.44	0.00	4.56	0.980	0.240	0.379
QTL HK	0.10	Uniform Strong	1.46	0.00	4.54	0.980	0.243	0.383
QTL IMP	0.10	Uniform Strong	1.32	0.00	4.68	0.960	0.220	0.351
BIC	0.10	Uniform Weak	3.58	2.62	2.42	0.612	0.597	0.592
EMS	0.10	Uniform Weak	4.34	2.74	1.66	0.645	0.723	0.668
QTL EM	0.10	Uniform Weak	1.14	0.00	4.86	0.920	0.190	0.310
QTL HK	0.10	Uniform Weak	1.08	0.00	4.92	0.880	0.180	0.294
QTL IMP	0.10	Uniform Weak	0.98	0.00	5.02	0.820	0.163	0.269
BIC	0.10	Random	5.10	2.32	0.90	0.720	0.850	0.771
EMS	0.10	Random	5.66	2.64	0.34	0.721	0.943	0.807
QTL EM	0.10	Random	1.38	0.00	4.62	0.940	0.230	0.363
QTL HK	0.10	Random	1.36	0.00	4.64	0.940	0.227	0.359
QTL IMP	0.10	Random	1.30	0.00	4.70	0.960	0.217	0.347

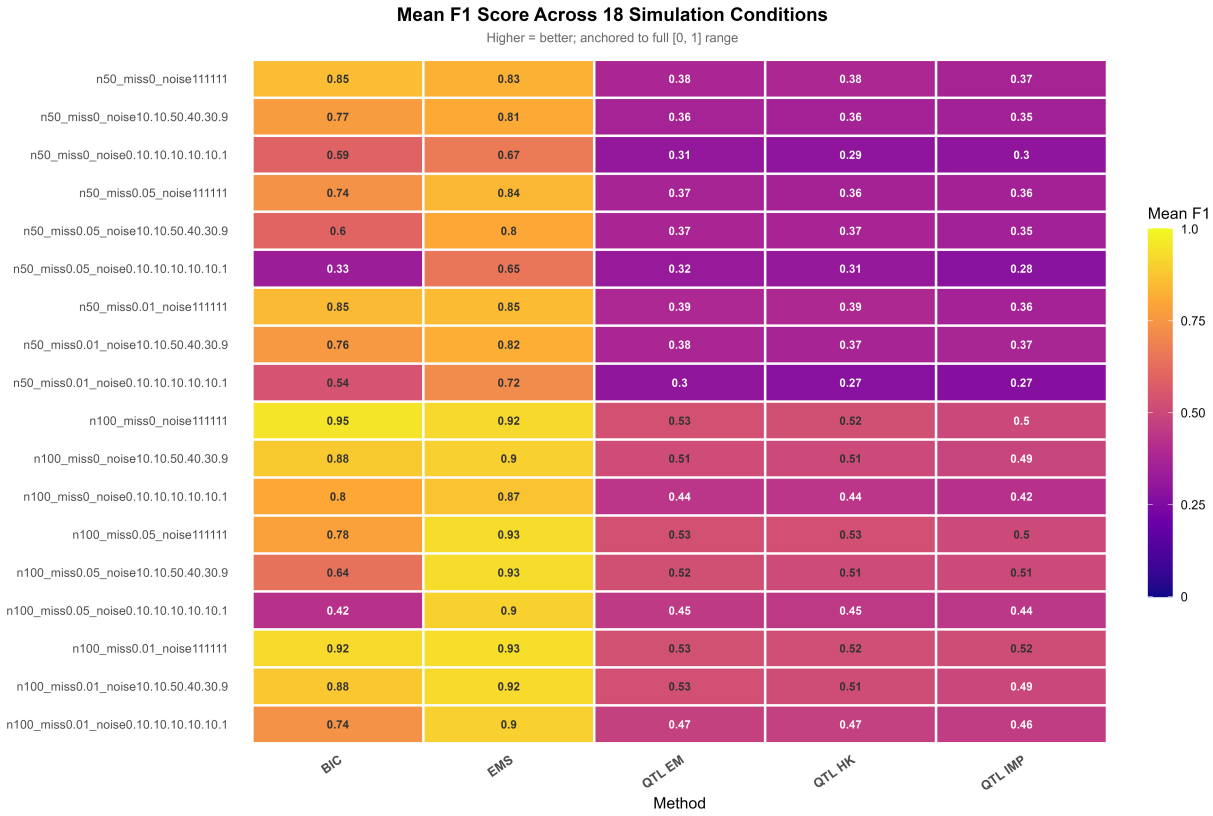

Figure 1: Mean F1 score across 18 simulation conditions. Higher values indicate better performance. Methods include Bayesian Information Criterion (BIC), Expectation-Maximization Selection (E-MS), and three QTL mapping approaches (EM, Haley-Knott, and multiple imputation) evaluated across uniform strong, uniform weak, and random missingness patterns at missingness rates of 0.01, 0.05, and naive 0.10 with sample sizes of n=50 and n=100.

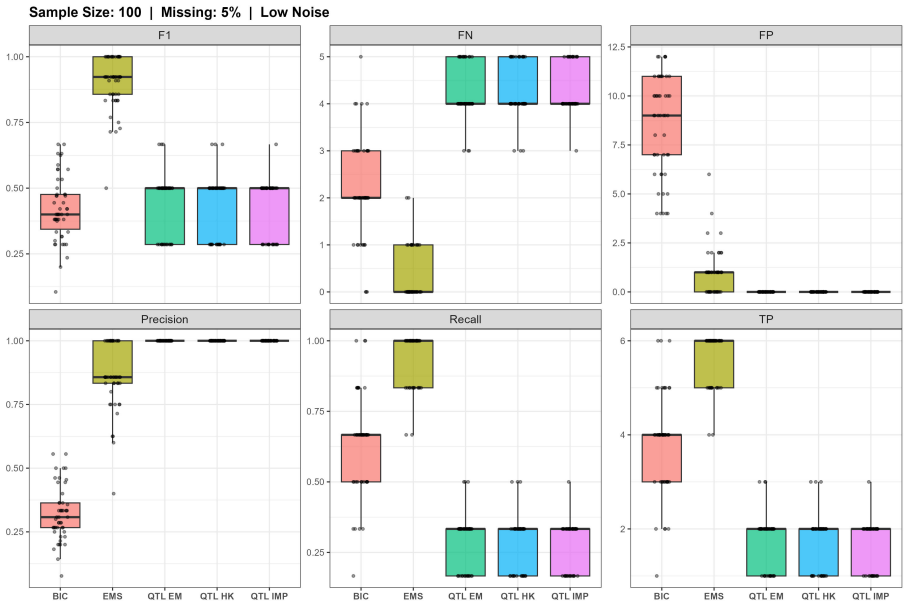

Figure 2: Performance across simulation condition 4. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under low noise conditions with 5% missing data (sample size: 100).

11 Appendix

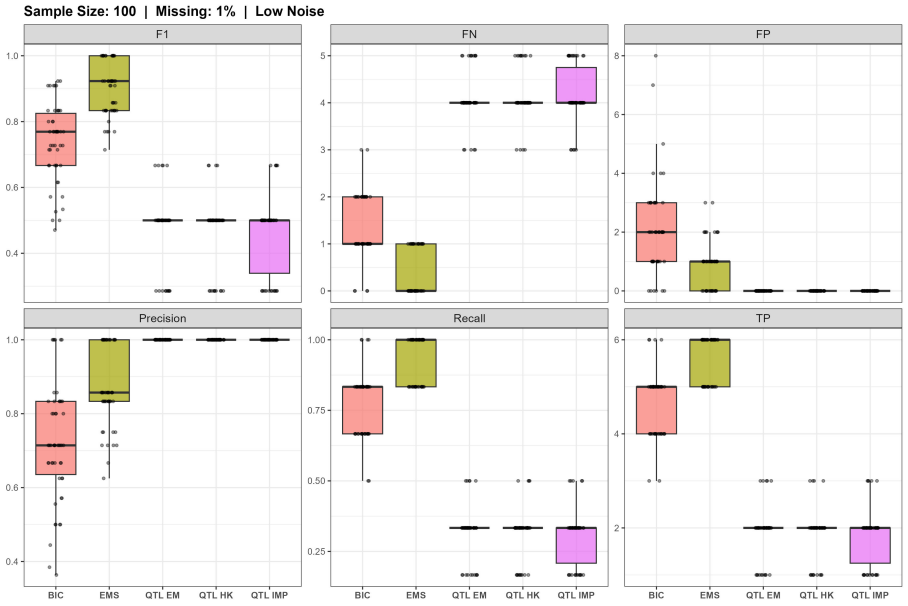

Figure 3: Performance across simulation condition 1. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under low noise conditions with 1% missing data (sample size: 100).

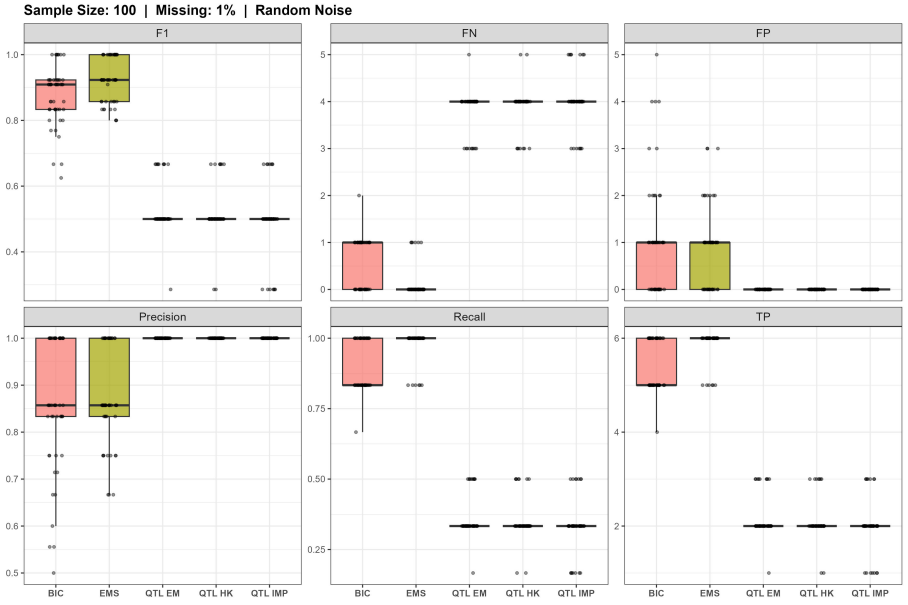

Figure 4: Performance across simulation condition 2. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under random noise conditions with 1% missing data (sample size: 100). QTL variants show more variable performance compared to EMS.

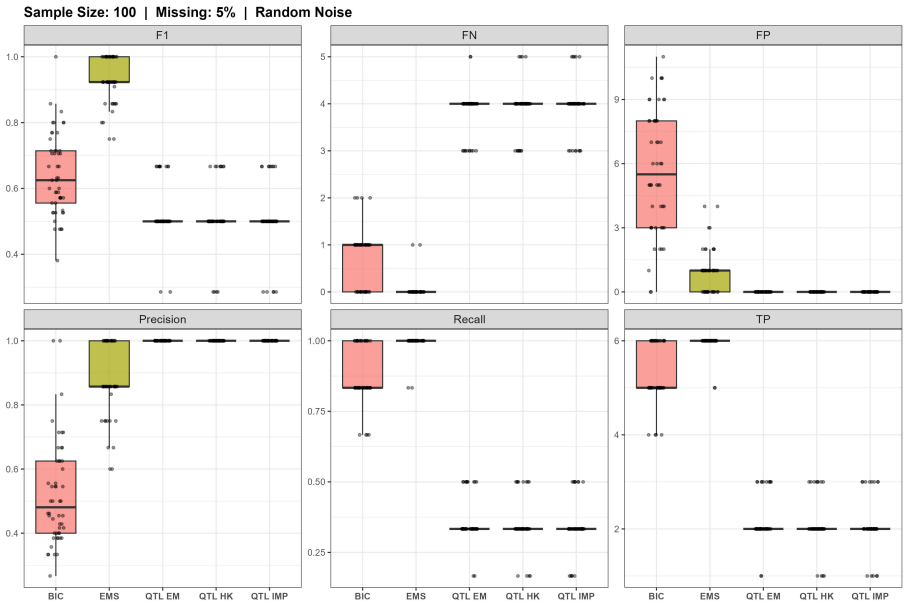

Figure 5: Performance across simulation condition 5. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under random noise conditions with 5% missing data (sample size: 100). EMS shows more consistent performance, while QTL EM exhibits greater variability.

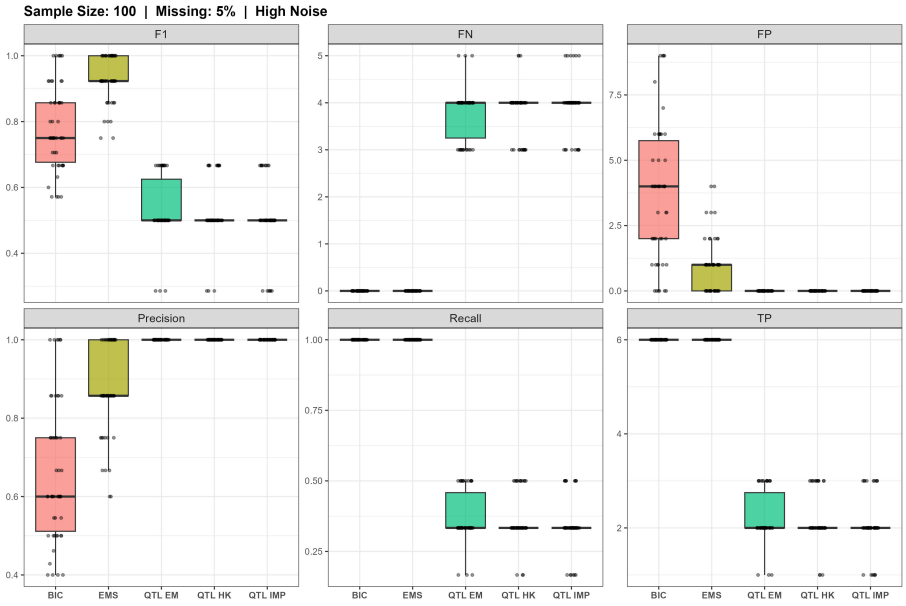

Figure 6: Performance across simulation condition 6. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under high noise conditions with 5% missing data (sample size: 100). QTL HK and QTL IMP maintain superior performance despite increased data missingness.

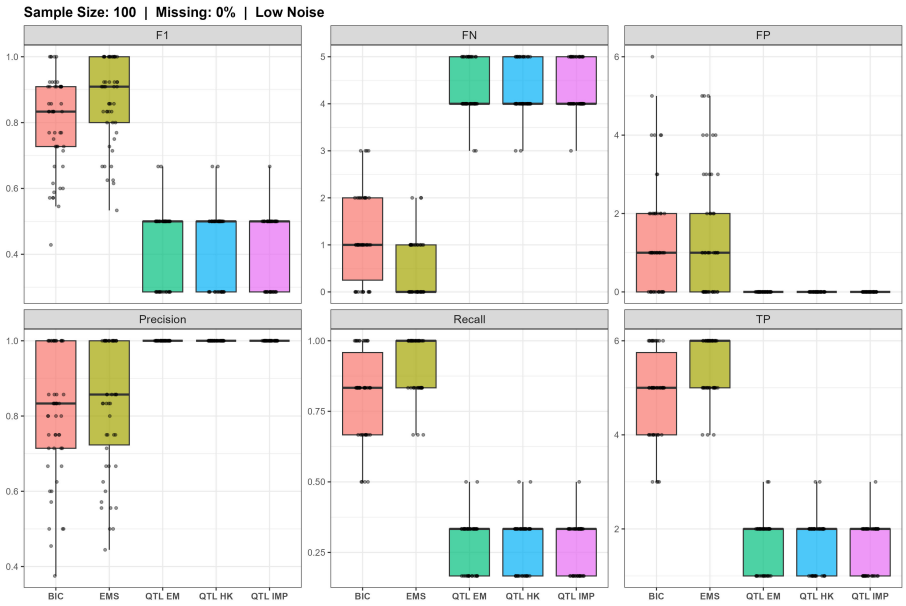

Figure 7: Performance across simulation condition 7. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under low noise conditions with 0% missing data (sample size: 100).

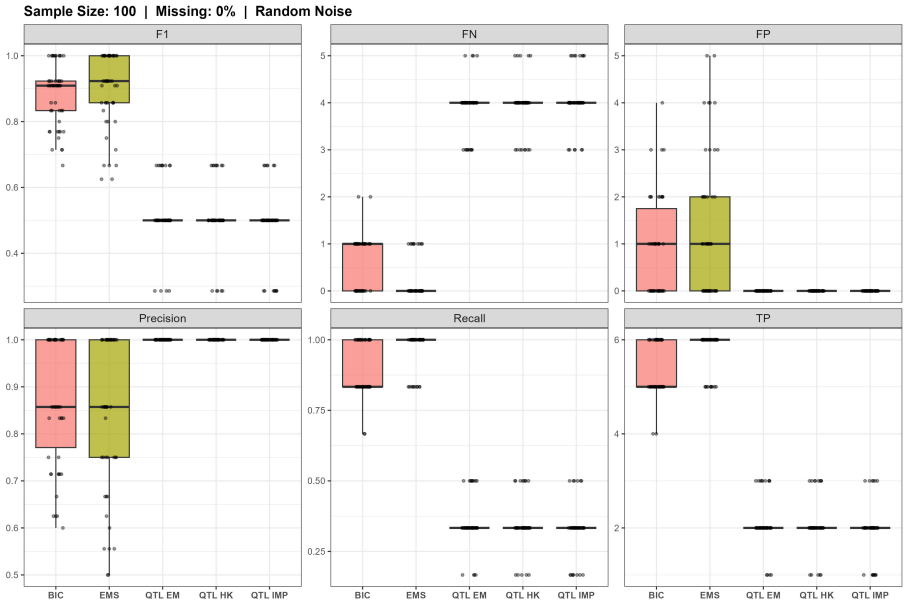

Figure 8: Performance across simulation condition 8. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under random noise conditions with 0% missing data (sample size: 100). Results show minimal variation across methods with complete data.

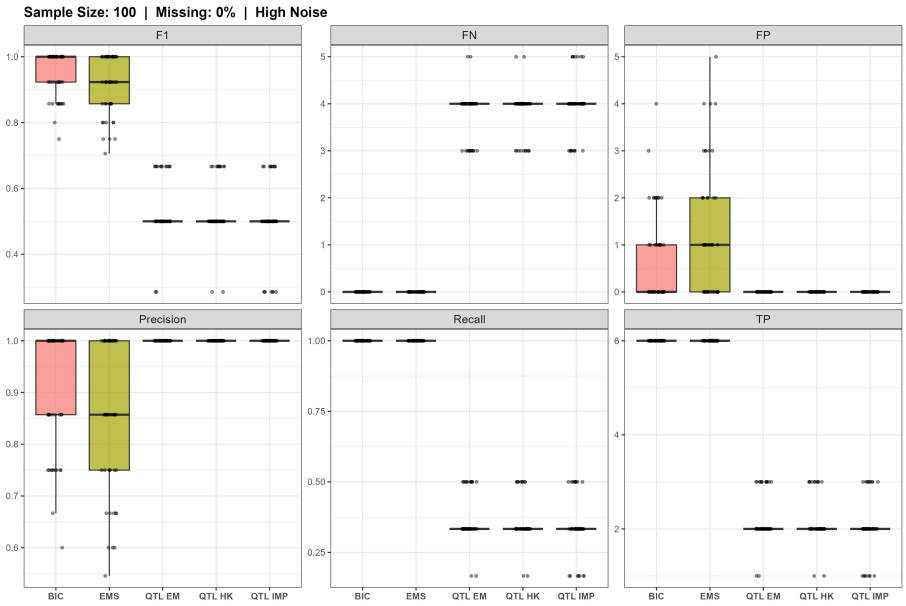

Figure 9: Performance across simulation condition 9. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under high noise conditions with 0% missing data (sample size: 100). QTL variants show enhanced stability in high noise even without missing data complications.

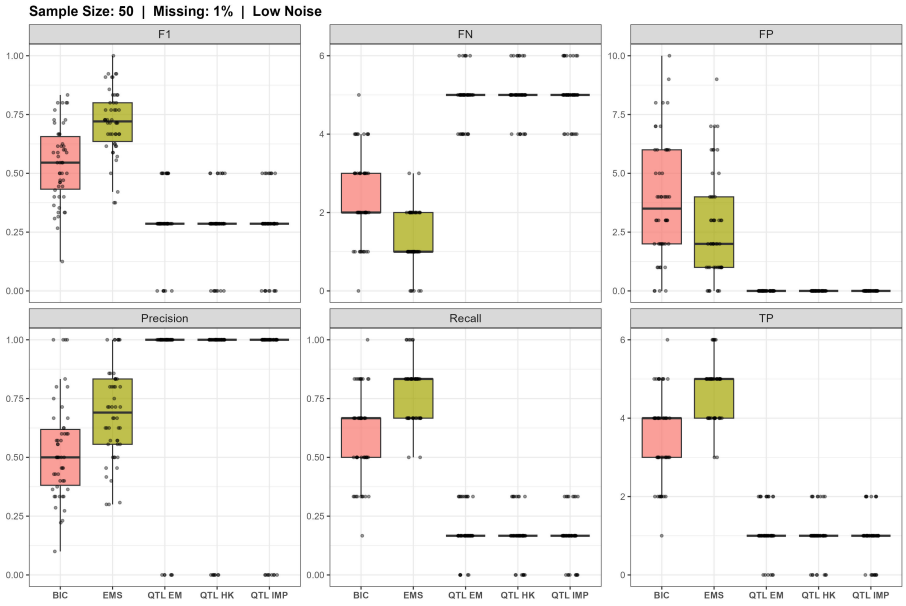

Figure 10: Performance across simulation condition 10. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under low noise conditions with 1% missing data (sample size: 50). Reduced sample size shows increased variability across all methods.

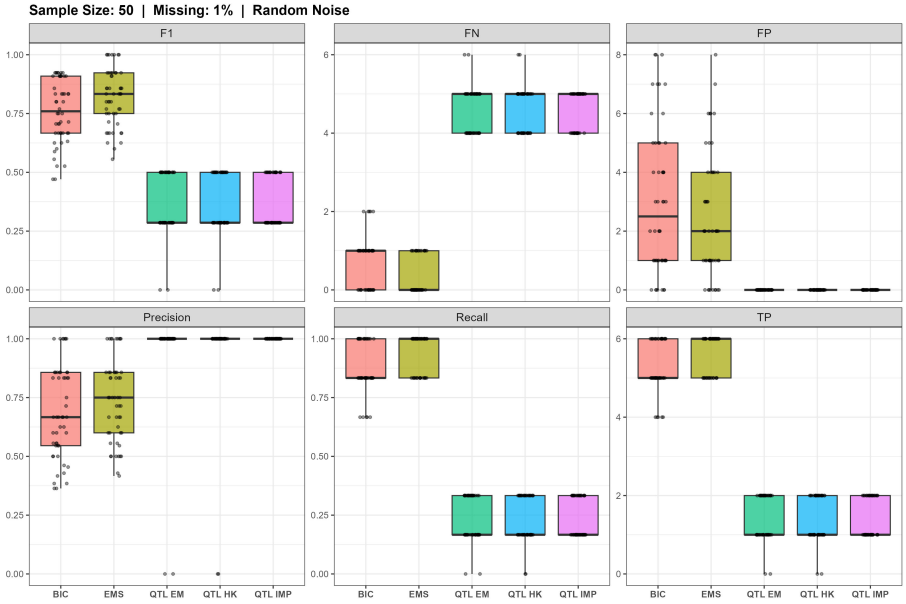

Figure 11: Performance across simulation condition 11. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under random noise conditions with 1% missing data (sample size: 50). Performance variance increases noticeably with smaller sample size.

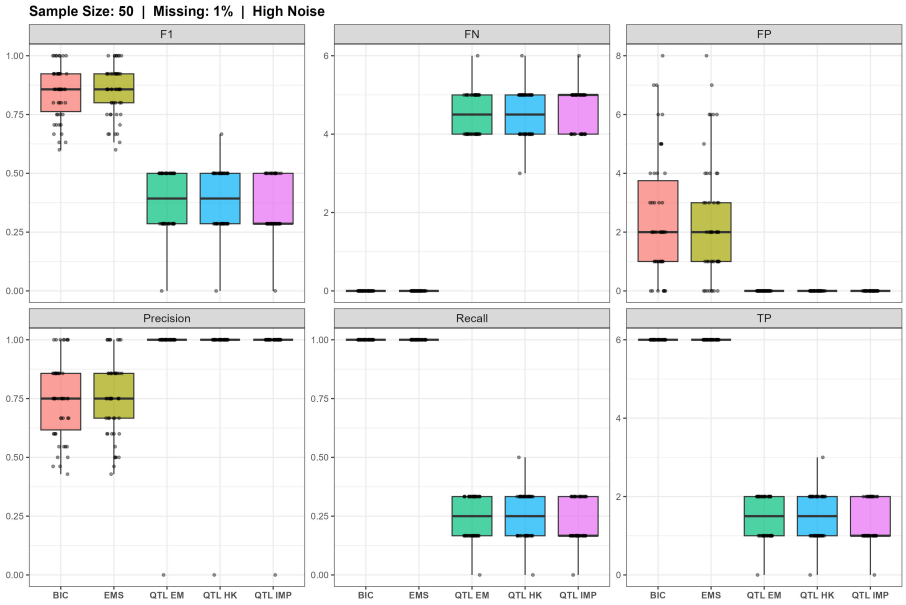

Figure 12: Performance across simulation condition 12. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under high noise conditions with 1% missing data (sample size: 50). QTL HK and QTL IMP continue to outperform baseline despite sample size reduction.

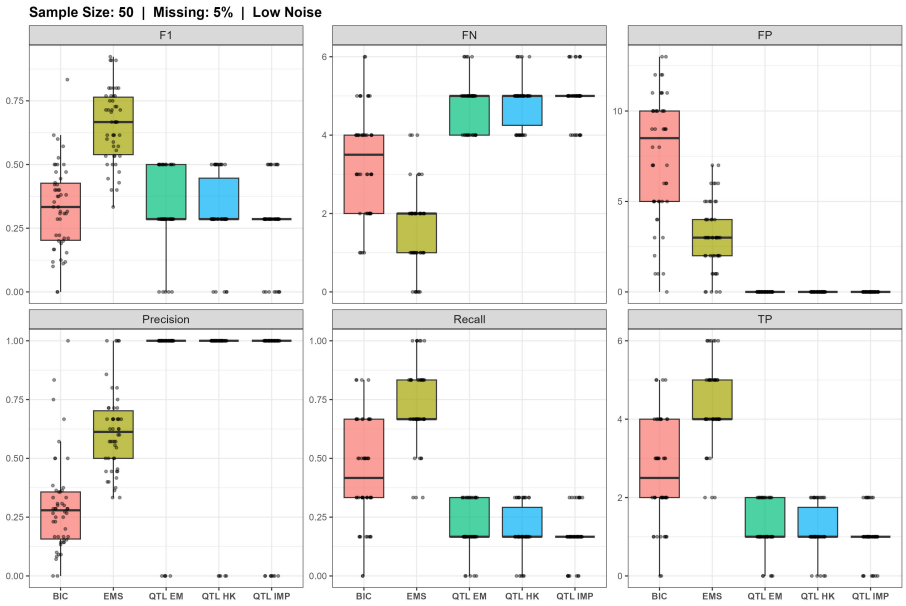

Figure 13: Performance across simulation condition 13. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under low noise conditions with 5% missing data (sample size: 50). Increased missingness combined with smaller sample yields greater performance dispersion.

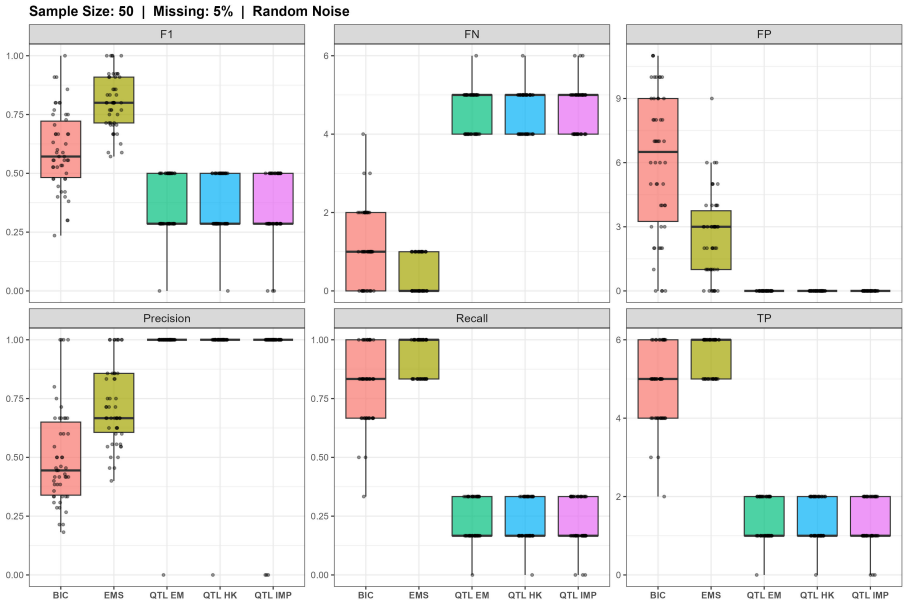

Figure 14: Performance across simulation condition 14. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under random noise conditions with 5% missing data (sample size: 50). EMS demonstrates more consistent behavior under these combined constraints.

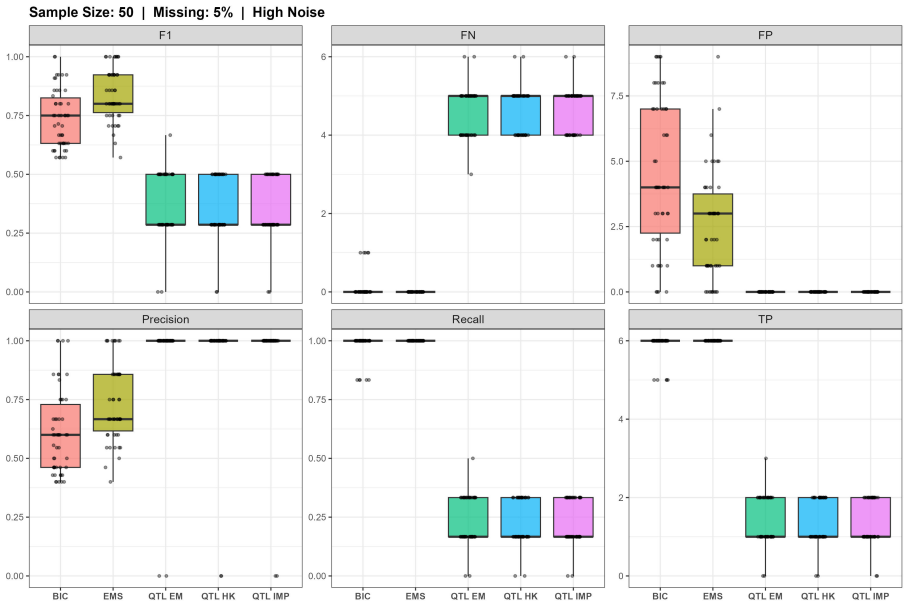

Figure 15: Performance across simulation condition 15. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under high noise conditions with 5% missing data (sample size: 50). QTL variants maintain advantages even under worst-case data conditions with limited sample size.

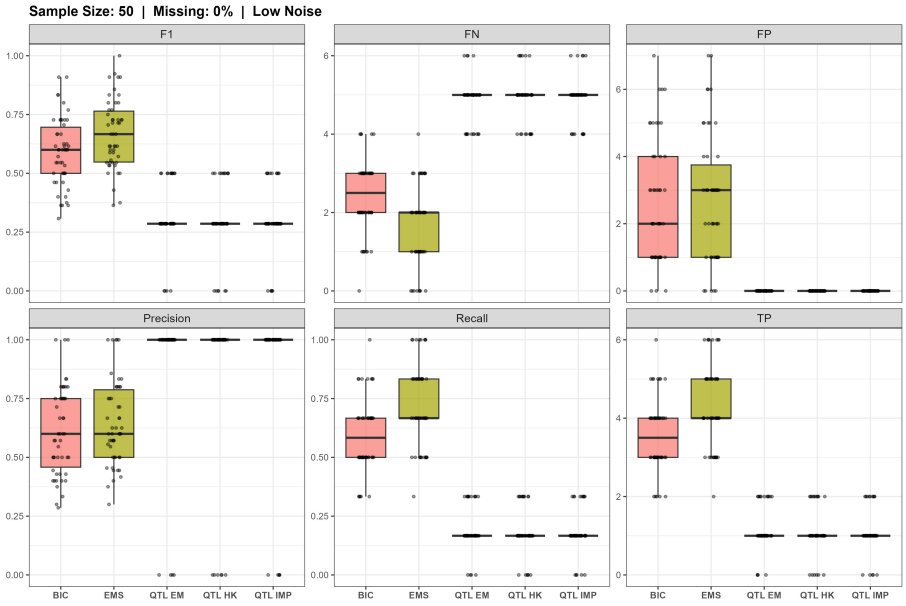

Figure 16: Performance across simulation condition 16. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under low noise conditions with 0% missing data (sample size: 50). Minimal missing data reduces variability despite smaller sample size.

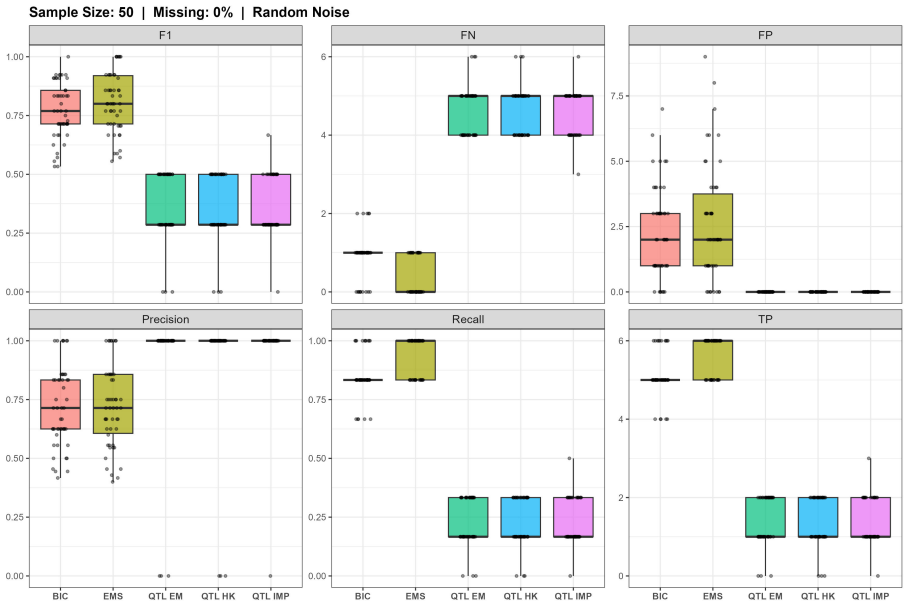

Figure 17: Performance across simulation condition 17. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under random noise conditions with 0% missing data (sample size: 50). Methods show remarkably consistent performance with complete data.

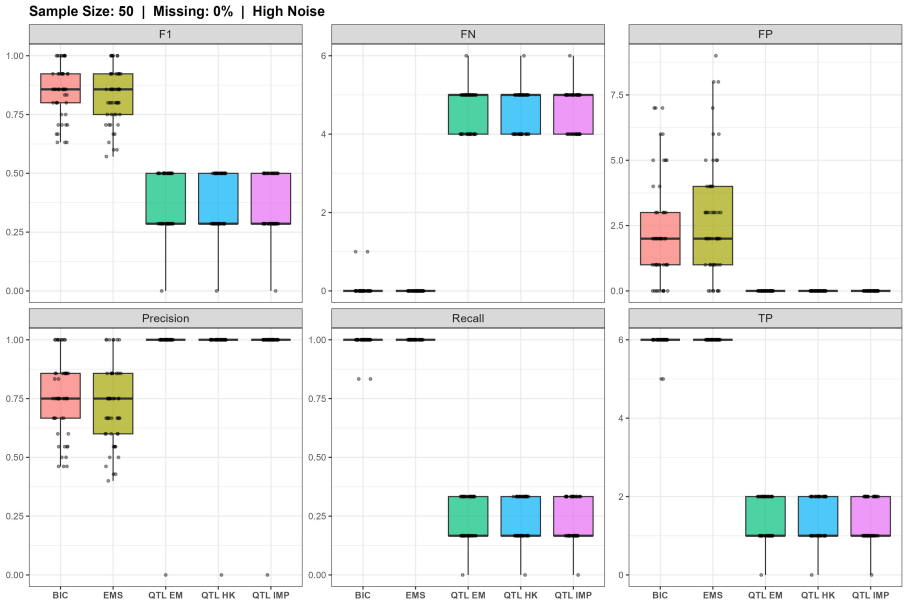

Figure 18: Performance across simulation condition 18. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under high noise conditions with 0% missing data (sample size: 50). Despite high noise, complete data allows all methods to maintain relatively stable performance trajectories.

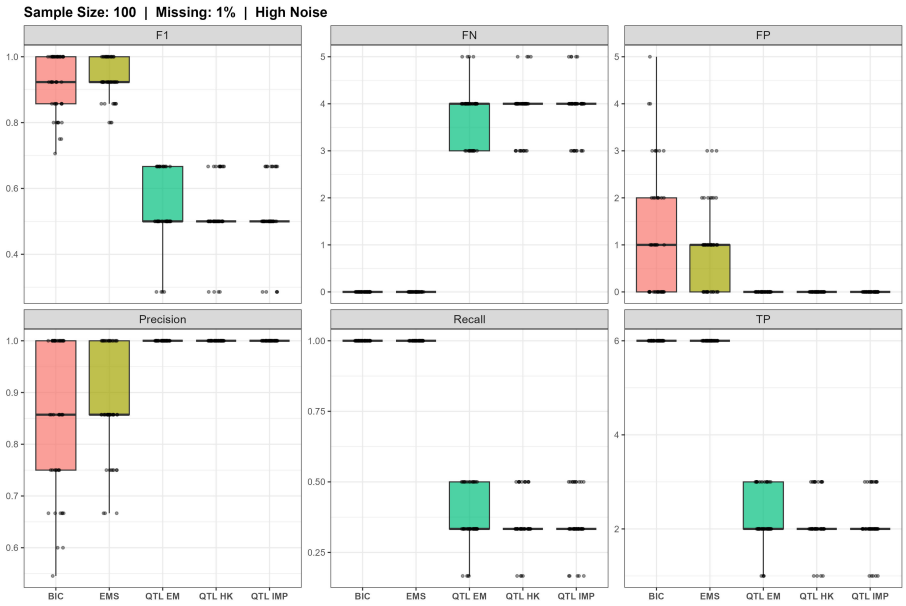

Figure 19: Performance across simulation condition 3. Boxplots comparing F1, FN, FP, Precision, Recall, and TP metrics across baseline (BIC) and four alternative methods (EMS, QTL EM, QTL HK, QTL IMP) under high noise conditions with 1% missing data (sample size: 100). QTL HK and QTL IMP demonstrate notably improved performance over BIC and EMS in high noise scenarios.